<u>EE 708:</u> Information Theory and Coding, Spring semester, 2025-26

<u>Slot:</u> 3 (Mon 10:30-11:25, Tue 11:30-12:25, Thu 8:30-9:30)
<u>Venue</u>: ~~EEG 401,~~ EEG 101 (from 29 January 2026)

Textbooks:

1. Elements of Information Theory – Cover and Thomas (2nd Ed)

Reference books:

1. Information Theory - Csiszar and Korner
2. Network Information Theory – Kim and El Gamal
3. Information Theory and Reliable Communication – R. Gallager

Semester Schedule:

Instruction begins:	05.01.2026 (Monday) - as per **IITB Calendar**
Quiz I	29.01.2026 (Thursday)
Attendance status intimation(*)	By Midsem
Midsem:	21.02.2026 (Saturday)-01.03.2026 (Sunday) - as per **IITB Calendar**
Quiz II	24.03.2026 (Tuesday)
Attendance status intimation(*)	14.03.2026 (Saturday)
Attendance status intimation(*)	14.04.2026 (Tuesday)
Award of DX grades:	15.04.2026 (Wednesday)-16.04.206 (Thursday) - as per **IITB Calendar**
Last day of instruction:	17.04.2026 (Friday) - as per **IITB Calendar**
Endsemester exam:	20.04.2026 (Monday)- 01.05.2026 (Friday) - as per **IITB Calendar**

(*) Attendance status for each student will be uploaded on moodle on the dates specified above

Grading:

1. Endsem : 45%

2. Midsem: 20%

3. Quiz 1: 13%

4. Quiz 2: 12%

5. Class participation: 10%

Attendance policy:

Attendance will be recorded. The marks on class participation will be based on attendance, including in the first and last weeks of classes. The marks (in class participation) will be assigned as follows:
If the attendance is at least 80%: full marks
If the attendance is n classes less than 80%: max{0, (10 - n)} marks

Exam attendance Quiz 1, Quiz 2, Midsem:

- Absence on the medical ground as certified by IIT Bombay hospital: Marks will be extrapolated from other exams
- Absence due to other unavoidable convincing reasons with proofs (only if permission is taken in writing beforehand): Marks will be extrapolated from other exams
- Absence due to other reasons: 0 marks for that exam

Important: In case of absence in an examination due to illness, the following are required.
(i) Send an email to the instructor, attaching a photo/scan of the medical certificate from IITB hospital
(ii) Show the original medical certificate from IITB hospital to the instructor
(iii) The medical certificate should say that you are advised to take rest at least for that day (day of the exam)

Both (i) and (ii) above should be done within 7 calendar days after the examination.

Endsem attendance:

Absence will result in FR grade
Re-exam only for medical reasons. As per UG rule book item 6.6: *"Serious illness/ accident in the case of the student herself/ himself. Serious illness/ accident/ death of parent/ guardian."*

Syllabus:

Introduction: Definition of information quantities with examples, log-sum inequality, Jensen's inequality, Fano's inequality, data processing inequality, sufficient statistics

Source coding techniques: Types of codes – singular, non-singular, uniquely decodable, prefix-free codes; Huffman code,Arithmetic codes, Lempel-Ziv codes,
Kraft inequality, Optimal codes, source coding theorem under zero error.

Typicallity: Typical sequences – weak and strong typicallity, AEP, Counting results on typical sequences, source coding theorem under zero error and small error using AEP.

Channel coding: Channels, capacity, and the channel coding theorem: Discrete memoryless channels, capacity calculations, jointly typical sequences and their counting, channel coding theorem with proof, average error versus maximum error – expurgation of codes, feedback does not improve capacity

Differential entropy, multivariate Gaussian distribution, Gaussian distribution maximizes differential entropy for a given covariance matrix

AWGN channels: Capacity calculation, entropy power inequality, parallel AWGN channels, power allocation – water pouring, Capacity of continuous time band-limited AWGN channels

Network information theory (depending on available time): Multiple access channels, source coding of distributed correlated sources, degraded broadcast channels and superposition coding

Rate-distortion theory (depending on available time)